



SCAFFOLD DESIGN REPORT

NIGERIA LIQUEFIED NATURAL GAS LIMITED



LOCATION: TRAIN 3	AREA: D
PERMIT TO WORK No.: -	TYPE OF SCAFFOLD: Composite
SCAFFOLD DRAWING No: PMS-WA-ARCO-034	
RISK CLASSIFICATION: ☒ High ☐ Medium ☐ Low	

REFERENCE STANDARDS

- BS EN 1991-1-4:2005 - Wind Actions
- BS EN 1993-1-1:2005 - Design of Steel Structures (Eurocode 3)
- BS EN 12811-1:2003 - Temporary Works Equipment - Scaffolds

SCAFFOLD TO BE INSPECTED & TAGGED BEFORE USE

BRIEF DESCRIPTION OF SCAFFOLD DESIGN

This document describes the structural design of a Composite scaffold required for scaffold access and maintenance activities for equipment tag 314LICA-013 under work order 314LICA-013 at TRAIN 3 / D. The scaffold envelope is 4m x 3m x 15m (length x width x height) and is configured with standards, ledgers, transoms, bracing, platforms, access, and support/tie arrangements as indicated on the approved drawing. The structure is 15m above ground level.

SAFE WORKING LOAD (SWL): 1.5 kN/m²

Note: This is a category 1 engineered scaffold. It must be inspected by the assurance team before use.

SITE / EQUIPMENT PHOTO

FILENAME	DATE/TIME	TYPE	SIZE
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TOTAL HORIZONTAL LOAD SUMMARY

Direction	Wind (kN)	Handrail (kN)	TOTAL (kN)
X	38.25	14.1	52.35
Z	34.75	9.6	44.35

Role	ID	Name	Signature	Date
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APPROVED BY				

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ERMT NO: PIC/PMS/26/2433

	Project Name:	SCAFFOLD DESIGN REPORT		
	Scaffold Type:	Composite	Calculation Sheet	
	Document Number:	PMS-WA-ARCO-034	Prepared By: Nnamani, Ifeanyi	

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GENERAL NOTE

BS EN 12811-1, 4.2.1.2 Scaffold materials are in accordance with BS EN 12811 with minimum design strength of 210 N/mm². This design calculation was carried out in accordance with the limit states design method stipulated in EN 1993-1-1:2005 (Eurocode 3 - Design of Steel Structures).

Material Properties

48.3 mm outside diameter x 3.6 mm thick steel tube was defined by STAAD.Pro Connect Edition V22 section tables (British) and assigned to all scaffold members as section **PIP483.6**.

Modelling Philosophy

BS EN 12811-1, 4.2.1.2 A 3-D model was analysed using **STAAD.Pro CONNECT Edition (V22)** in compliance with EN 1993-1-1:2005. Base supports are modelled as pinned with lateral spring stiffness representing frictional restraint ($K_{FX}/K_{FZ} = 21.112 \text{ kN}$). Base supports are modelled as elastic spring supports, restrained laterally in both horizontal directions with rotational degrees of freedom released, representing flexible ground contact conditions at each standard foot. See [Appendix A](#) for model views.

LOAD CASES

Load diagrams are shown in [Appendix B](#).

Dead Load / Self-weight Self-weight of structural elements computed by STAAD.Pro with steel density **76.8195 kN/m³**, with a **10% allowance** for fittings and couplers (selfweight factor = 1.1).

Live Load on Platforms

BS EN 12811-1 Platform live loading was applied to the scaffold working platforms as member UDLs in the STAAD model. The platform load intensity is multiplied by the tributary width to obtain the line load applied to platform supporting members. (LC 2).

Load Class	Load Intensity (kN/m ²)	Tributary Width (m)	Applied STAAD Line Load (kN/m)	Working
2	1.5	1.000 / 2 = 0.500	0.75	1.5 x 0.5 = 0.75

Governing platform live load intensity used for SWL = **1.5 kN/m²**.
Total applied platform live load from STAAD output = **132.0 kN**.

Handrail Load

BS EN 12811-1 Handrail loads were applied where required by the scaffold configuration and modelled independently in the global X and Z directions.

Handrail Load (HLX) = **14.1 kN** (member UDL 0.15 kN/m)
Handrail Load (HLZ) = **9.6 kN** (member UDL 0.15 kN/m)

Wind Load

EN 1991-1-4:2005 Wind loads were derived in accordance with EN 1991-1-4:2005 using NLNG Bonny Island site parameters. Full calculation chain is given in the Wind Load Calculation sheet below. Wind UDL = **0.073468 kN/m** applied as a member UDL to all exposed scaffold members in the global X and Z wind directions.

Total Wind Load - X Axis (WLX) = **38.25 kN**
 Total Wind Load - Z Axis (WLZ) = **34.75 kN**

WIND LOAD CALCULATION - EN 1991-1-4:2005

Reference	Calculation	Value	Unit
	Basic wind speed (3-second gust)		
	V3 =	36.0	m/s
	<i>Eurocode $V_{b,0}$ is the characteristic 10-minute mean wind velocity</i>		
Dredger P. - How to Approximate	Convert 3-second gust to 10-minute mean wind velocity		
	V3 / V3600 =	1.52	
	V600 / V3600 =	1.06	
	V600 = (V600/V3600) x (V3600/V3) x V3 =	25.11	m/s
	Basic wind speed $V_b = V600 =$	25.11	m/s
EN 1991-1-4:2005	Roughness Length (z0) - Terrain Category 0	0.003	m
	Terrain category II (z0,ii)	0.05	
EN 1991-1-4, Table 4.1	Z _{min}	1.0	m
	z - height of structure under consideration	15.0	m
EN 1991-1-4, Sec 4.3	C _{r0} (z) - Roughness factor	1.0	
EN 1991-1-4, Sec 4.3	C _o (z) - Orography factor	1.0	
EN 1991-1-4, Sec 4.3	Terrain factor K _r = 0.19 x [z0/z0,ii] ^{0.07}	0.156	
EN 1991-1-4, Table 4.1	Z _{min} < z -> Case 1 applies		
	Wind Turbulence		
	C _r (z) = K _r x ln(z/z0)	1.329	
	Mean velocity V _m (z) = C _r (z) x C _o (z) x V _b	33.365	m/s
EN 1991-1-4, Sec 4.4	K1	1.0	
	Turbulence intensity I _v (z) = k1 / [C _o (z) x ln(z/z0)]	0.1174	
	Peak Velocity Pressure		
	Density of air rho	1.25	kg/m³
EN 1991-1-4, Sec 4.5	q _p (z) = [1 + 7*I _v (z)] x 0.5 x rho x V _m (z)²	1267.5573	N/m²
Wind Load = C_f x q_p x A_f			
C _f =		1.2	
q _p =		1.267557	kN/m²
A _f =	Tube outer diameter (48.3 mm)	0.0483	m
Wind Load UDL =		0.073468	kN/m

LOAD COMBINATIONS

BS EN 12811-1, 6.2.9.2

Load combinations in accordance with BS EN 12811-1:2003. ULS combinations use gamma = **1.5**;
 SLS (characteristic) combinations use gamma = **1.0**.

DL = Dead Load | **LL** = Platform Live Load | **HLX/HLZ** = Handrail X/Z | **WLX/WLZ** = Wind X/Z

LC No.	Combination	Component Loadings & Factors
7	DL + LL	LC1 x 1.0 + LC2 x 1.0
8	DL + HLX	LC1 x 1.0 + LC3 x 1.0
9	DL + HLZ	LC1 x 1.0 + LC4 x 1.0

LC No.	Combination	Component Loadings & Factors
10	DL + WLX	LC1 x 1.0 + LC5 x 1.0
11	DL + WLZ	LC1 x 1.0 + LC6 x 1.0
12	DL + LL + HLX	LC1 x 1.0 + LC2 x 1.0 + LC3 x 1.0
13	DL + LL + HLZ	LC1 x 1.0 + LC2 x 1.0 + LC4 x 1.0
14	DL + LL + WLX	LC1 x 1.0 + LC2 x 1.0 + LC5 x 1.0
15	DL + LL + WLZ	LC1 x 1.0 + LC2 x 1.0 + LC6 x 1.0
16	DL + LL + HLX + WLX	LC1 x 1.0 + LC2 x 1.0 + LC3 x 1.0 + LC5 x 1.0
17	DL + LL + HLZ + WLZ	LC1 x 1.0 + LC2 x 1.0 + LC4 x 1.0 + LC6 x 1.0
18	DL + LL + HLX + WLZ	LC1 x 1.0 + LC2 x 1.0 + LC3 x 1.0 + LC6 x 1.0
19	DL + LL + HLZ + WLX	LC1 x 1.0 + LC2 x 1.0 + LC4 x 1.0 + LC5 x 1.0
20	1.5DL	LC1 x 1.5
21	1.5DL + 1.5LL	LC1 x 1.5 + LC2 x 1.5
22	1.5DL + 1.5HLX	LC1 x 1.5 + LC3 x 1.5
23	1.5DL + 1.5HLZ	LC1 x 1.5 + LC4 x 1.5
24	1.5DL + 1.5WLX	LC1 x 1.5 + LC5 x 1.5
25	1.5DL + 1.5WLZ	LC1 x 1.5 + LC6 x 1.5
26	1.5DL + 1.5LL + 1.5HLX	LC1 x 1.5 + LC2 x 1.5 + LC3 x 1.5
27	1.5DL + 1.5LL + 1.5HLZ	LC1 x 1.5 + LC2 x 1.5 + LC4 x 1.5
28	1.5DL + 1.5LL + 1.5WLX	LC1 x 1.5 + LC2 x 1.5 + LC5 x 1.5
29	1.5DL + 1.5LL + 1.5WLZ	LC1 x 1.5 + LC2 x 1.5 + LC6 x 1.5
30	1.5DL + 1.5LL + 1.5HLX + 1.5WLX	LC1 x 1.5 + LC2 x 1.5 + LC3 x 1.5 + LC5 x 1.5
31	1.5DL + 1.5LL + 1.5HLZ + 1.5WLZ	LC1 x 1.5 + LC2 x 1.5 + LC4 x 1.5 + LC6 x 1.5
32	1.5DL + 1.5LL + 1.5HLX + 1.5WLZ	LC1 x 1.5 + LC2 x 1.5 + LC3 x 1.5 + LC6 x 1.5
33	1.5DL + 1.5LL + 1.5HLZ + 1.5WLX	LC1 x 1.5 + LC2 x 1.5 + LC4 x 1.5 + LC5 x 1.5

STATIC CHECK PARAMETERS - UTILIZATION RATIO SUMMARY

Maximum Utilization Ratio = 0.653 at Member No. **836**, Load Combination **30**, Clause **EC-6.3.3-662**. **PASS** - All members adequate; UC ratio below unity.

UC ratio diagram: see [Appendix C](#).

Member	Section	Status	Clause	UC Ratio	L/C	F _x (kN)	Type
836	PIP483.6	PASS	EC-6.3.3-662	0.653	30	0.14	C
837	PIP483.6	PASS	EC-6.3.3-662	0.652	30	0.14	C
716	PIP483.6	PASS	EC-6.2.9.1	0.579	30	0.00	C
717	PIP483.6	PASS	EC-6.2.9.1	0.578	30	0.00	C
401	PIP483.6	PASS	EC-6.2.7(5)	0.567	30	8.34	C
399	PIP483.6	PASS	EC-6.2.7(5)	0.566	30	8.33	C
777	PIP483.6	PASS	EC-6.3.3-662	0.463	30	2.61	C
775	PIP483.6	PASS	EC-6.3.3-662	0.462	30	2.60	C
386	PIP483.6	PASS	EC-6.3.3-662	0.420	30	5.31	C
388	PIP483.6	PASS	EC-6.3.3-662	0.420	30	5.30	C
396	PIP483.6	PASS	EC-6.2.9.1	0.418	30	0.74	C
398	PIP483.6	PASS	EC-6.2.9.1	0.417	30	0.74	C
772	PIP483.6	PASS	EC-6.3.3-662	0.413	30	1.51	C
774	PIP483.6	PASS	EC-6.3.3-662	0.413	30	1.51	C
395	PIP483.6	PASS	EC-6.2.9.1	0.406	30	2.19	C
393	PIP483.6	PASS	EC-6.2.9.1	0.405	30	2.18	C

Member	Section	Status	Clause	UC Ratio	L/C	F _x (kN)	Type
833	PIP483.6	PASS	EC-6.2.9.1	0.403	31	0.20	T
834	PIP483.6	PASS	EC-6.2.9.1	0.397	31	0.20	T
835	PIP483.6	PASS	EC-6.3.3-662	0.393	31	0.54	C
832	PIP483.6	PASS	EC-6.3.3-662	0.386	31	0.55	C
391	PIP483.6	PASS	EC-6.3.3-662	0.371	31	3.43	C
390	PIP483.6	PASS	EC-6.3.3-662	0.370	31	3.42	C
776	PIP483.6	PASS	EC-6.3.3-662	0.364	31	1.54	C
773	PIP483.6	PASS	EC-6.3.3-662	0.361	31	1.40	C
655	PIP483.6	PASS	EC-6.3.3-662	0.352	30	5.60	C
657	PIP483.6	PASS	EC-6.3.3-662	0.352	30	5.61	C
97	PIP483.6	PASS	EC-6.3.3-662	0.350	31	15.64	C
760	PIP483.6	PASS	EC-6.3.3-662	0.345	30	1.31	C
766	PIP483.6	PASS	EC-6.3.3-662	0.345	30	1.31	C
93	PIP483.6	PASS	EC-6.3.3-662	0.344	33	16.20	C
96	PIP483.6	PASS	EC-6.3.3-662	0.338	30	14.79	C
98	PIP483.6	PASS	EC-6.3.3-662	0.337	31	14.65	C
381	PIP483.6	PASS	EC-6.3.3-662	0.332	31	0.63	C
713	PIP483.6	PASS	EC-6.3.3-662	0.331	31	0.14	C
376	PIP483.6	PASS	EC-6.2.9.1	0.330	31	0.02	C
379	PIP483.6	PASS	EC-6.2.9.1	0.330	31	0.36	T
818	PIP483.6	PASS	EC-6.3.3-662	0.330	30	0.52	C
827	PIP483.6	PASS	EC-6.3.3-662	0.330	30	0.52	C
374	PIP483.6	PASS	EC-6.3.3-662	0.327	31	0.16	C
715	PIP483.6	PASS	EC-6.2.9.1	0.327	31	0.36	T
95	PIP483.6	PASS	EC-6.3.3-662	0.326	28	14.74	C
714	PIP483.6	PASS	EC-6.3.3-662	0.325	31	0.14	C
656	PIP483.6	PASS	EC-6.3.3-662	0.324	31	2.92	C
653	PIP483.6	PASS	EC-6.2.9.1	0.323	31	2.74	C
190	PIP483.6	PASS	EC-6.3.3-662	0.322	31	10.05	C
249	PIP483.6	PASS	EC-6.3.3-662	0.321	31	10.87	C
712	PIP483.6	PASS	EC-6.2.9.1	0.320	31	0.37	T
820	PIP483.6	PASS	EC-6.2.9.1	0.320	30	0.52	T
825	PIP483.6	PASS	EC-6.2.9.1	0.320	30	0.52	T
99	PIP483.6	PASS	EC-6.3.3-662	0.318	33	13.54	C

Showing highest 50 utilization ratios; 727 lower-utilization members omitted.

SCAFFOLD TUBES CONNECTION STABILITY CHECK

EN 12811, Table C.1

The maximum axial load on scaffold tube members (from STAAD.Pro Connect Edition analysis) is **16.550 kN** (Member 43, C). The required right-angle coupler class is selected from Table C.1 below using the slipping resistance limit.

Selected coupler class: **Review required** (capacity = 15.0 kN).

Check: 16.550 kN ≤ 15.0 kN → **FAIL - Review coupler arrangement**

Applied axial load exceeds the Class B slipping resistance in Table C.1.

Member force table: see [Appendix D](#).

Table C.1 - Characteristic Values of Resistances for Couplers (EN 12811)

Coupler Type	Resistance	Characteristic Value	
		Class A	Class B
Right-angle Coupler (RA)	Slipping force $F_{s,k}$ (kN)	10.0	15.0
	Cruciform bending moment $M_{B,k}$ (kNm)	-	0.8
	Pull-apart force $F_{p,k}$ (kN)	20.0	30.0
	Rotational moment $M_{T,k}$ (kNm)	-	0.13
Swivel Coupler (SW)	Slipping force $F_{s,k}$ (kN)	6.0	9.0
	Cruciform bending moment $M_{B,k}$ (kNm)	-	0.8
	Pull-apart force $F_{p,k}$ (kN)	15.0	22.5
	Rotational moment $M_{T,k}$ (kNm)	-	0.08
Putlog / Single Coupler (PL)	Slipping force $F_{s,k}$ (kN)	-	3.0
	Cruciform bending moment $M_{B,k}$ (kNm)	-	0.6

FRICTIONAL RESISTANCE

Max vertical reaction per support ($\text{Max } f_y$) = Resistance x 0.15 / C_F = **10.556 kN**

Coefficient of friction (C_F) = **0.3**

Frictional Resistance (FR) = C_F x Max f_y = **3.1668 kN**

Spring Stiffness = FR / 0.15 = **21.112 kN** <- applied as KFX/KFZ in STAAD model

DEFLECTION CHECK RESULTS

BS EN 12811-1, Cl 6.3 Serviceability limit state - values from STAAD.Pro SLS (unfactored) combination results. Diagrams and displacement tables are in [Appendix C](#) and [Appendix D](#).

Vertical Deflection Check

Parameter	Value	Allowable	Check
Max vertical displacement - LC 16 L = scaffold width = 3000 mm (default)	1.076 mm	$L/100 = 3000/100 = 30.0 \text{ mm}$	PASS

Horizontal Deflection Check

Parameter	Value	Allowable	Check
Max horizontal displacement - LC 16 L = scaffold depth = 4000 mm (default)	14.114 mm	$L/200 = 4000/200 = 20.0 \text{ mm}$	PASS

CONCLUSION

The structural analysis and design of the **Composite** scaffold confirms the structure is adequate for its intended purpose. All member utilization ratios are below unity (maximum UC = **0.653** at Member 836). Vertical and horizontal deflections are within their respective allowable limits of **$L/100 = 30.0\text{mm}$** and **$L/200 = 20.0\text{mm}$** . All scaffold tube coupler connections are adequate for the applied axial loads.

RECOMMENDATIONS

The scaffold shall be inspected by a competent Scaffold Inspector after erection to confirm conformity with the design specification and approved working drawings before use. All erection work shall comply with the design intent and

material specification. Any proposed change to the design requires a re-design and the Scaffold Engineer shall be duly informed.

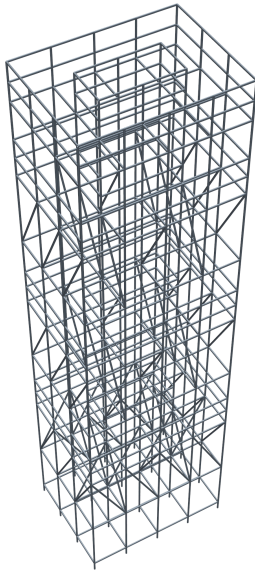
INSTALLATION NOTES

1. The scaffold working drawings and purpose of the scaffold must be reviewed and discussed by all personnel involved prior to execution.
2. Confirm that the Composite scaffold configuration matches the approved drawing and is suitable for scaffold access and maintenance activities.
3. The work vicinity must be barricaded and appropriate signage placed at all access points.
4. All scaffolders must wear complete PPE in accordance with NLNG work-at-height regulations.
5. A valid work permit must be obtained, discussed, and signed before erection commences.
6. Scaffold erection must be carried out in strict accordance with NASC guidance and site safety procedures.
7. Set out each scaffold zone shown on the composite drawing and confirm interface points between the combined scaffold types before erection proceeds.
8. Confirm the scaffold footprint/envelope is approximately 3.0m x 4.0m x 15.0m or as otherwise dimensioned on the approved drawing.
9. Erect scaffold tube standards (48.3 x 3.6 mm), ledgers, and transoms progressively from the base, checking plumb and level in both axes before proceeding to the next lift.
10. Install the lower horizontal ledgers and transoms at the kicker lift height (0.300m) to form the first stable frame.
11. Install mid-level horizontal ledgers and transoms at 2.0m height in accordance with the working drawing.
12. Install mid-level horizontal ledgers and transoms at 4.0m height in accordance with the working drawing.
13. Install mid-level horizontal ledgers and transoms at 4.5m height in accordance with the working drawing.
14. Install mid-level horizontal ledgers and transoms at 5.0m height in accordance with the working drawing.
15. Install mid-level horizontal ledgers and transoms at 6.0m height in accordance with the working drawing.
16. Install mid-level horizontal ledgers and transoms at 8.0m height in accordance with the working drawing.
17. Install mid-level horizontal ledgers and transoms at 8.5m height in accordance with the working drawing.
18. Install mid-level horizontal ledgers and transoms at 9.0m height in accordance with the working drawing.
19. Install mid-level horizontal ledgers and transoms at 10.0m height in accordance with the working drawing.
20. Install mid-level horizontal ledgers and transoms at 12.0m height in accordance with the working drawing.
21. Install mid-level horizontal ledgers and transoms at 12.5m height in accordance with the working drawing.
22. Install mid-level horizontal ledgers and transoms at 13.0m height in accordance with the working drawing.
23. Install mid-level horizontal ledgers and transoms at 14.0m height in accordance with the working drawing.
24. Install mid-level horizontal ledgers and transoms at 14.5m height in accordance with the working drawing.
25. Install all plan, face, and longitudinal bracing shown on the approved drawing to provide sway resistance in both principal directions.
26. Install platform boards, guardrails, mid-rails, toe boards, access arrangements, and any handrail load-resisting members required by the design.
27. Tighten and torque-check all load-bearing couplers and support connections before the scaffold is released for inspection.
28. Gravlock or box-tie the scaffold structure to the existing structure at all indicated tie points at 4.0m and 8.0m and 12.0m height as shown on the working drawing.

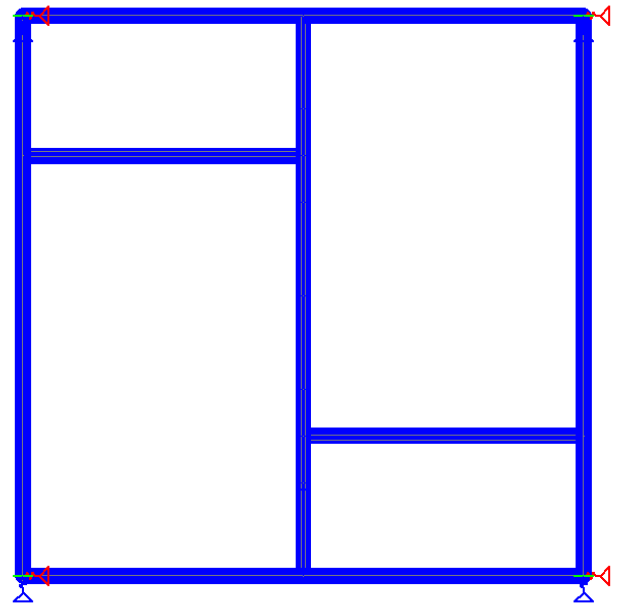
29. The scaffold structure must be inspected and tagged by a competent advance scaffold inspector after erection and before use.
30. The safe working load (SWL) stated on the design report must not be exceeded at any time during operations.
31. All modifications made after erection must be carried out by a competent scaffold team with the approval of the scaffold engineer.
End-user or third-party modification of this scaffold structure is strictly prohibited.
32. For dismantling, perform the erection steps in reverse order.

APPENDIX A - MODEL VIEWS

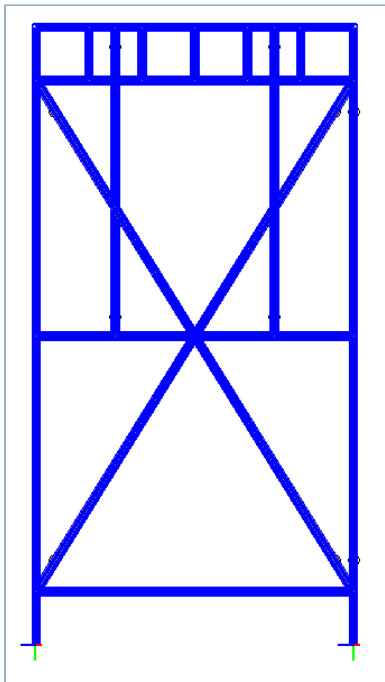
3D Isometric View



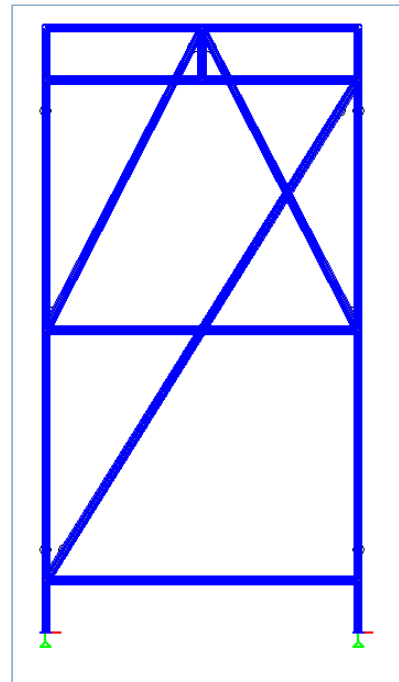
Plan View



Elevation from +X Axis



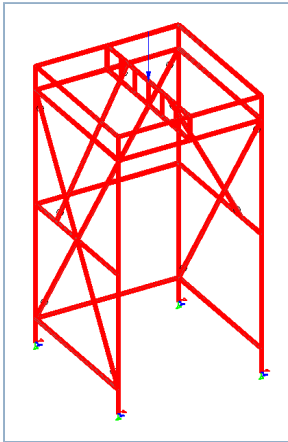
Elevation from +Z Axis



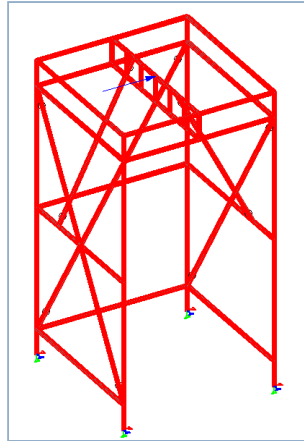
Scaffold Dimensions: W = 3.0m | D = 4.0m | H = 15.0m | **Lifts:** 0.3m, 2.0m, 4.0m, 4.5m, 5.0m, 6.0m, 8.0m, 8.5m, 9.0m, 10.0m, 12.0m, 12.5m, 13.0m, 14.0m, 14.5m, 15.0m

APPENDIX B - LOAD DIAGRAMS

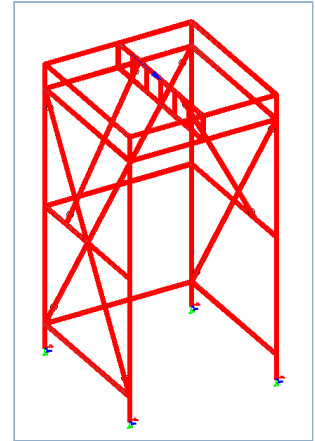
Live Load - Platform Live Load (LL)



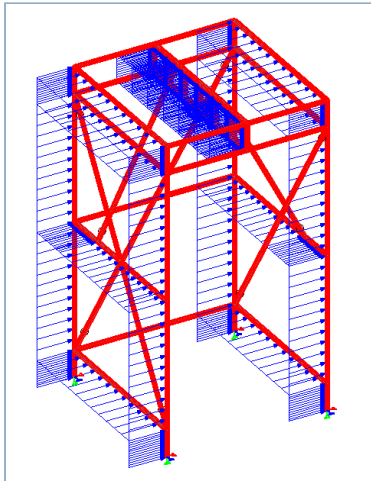
Handrail Load in X Direction (HLX)



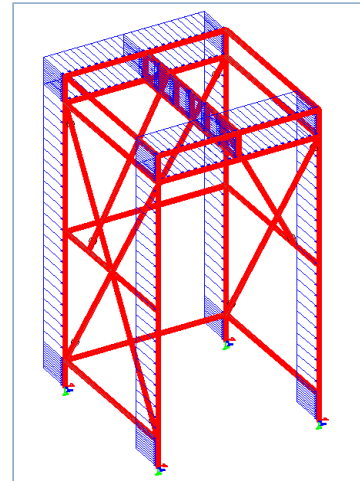
Handrail Load in Z Direction (HLZ)



Wind Load - X Axis (WLX)

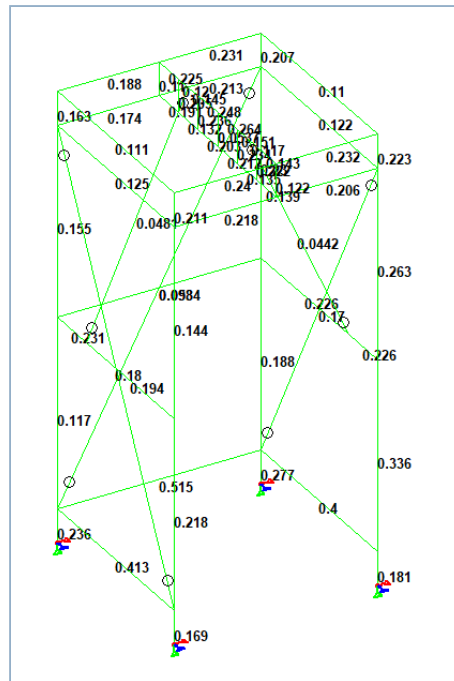


Wind Load - Z Axis (WLZ)

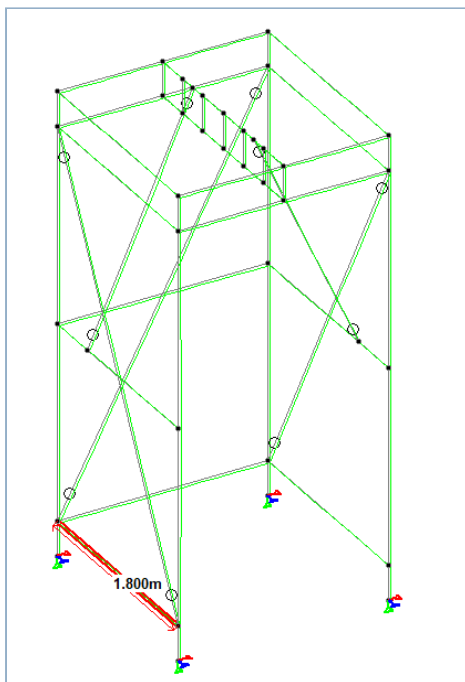


APPENDIX C - ANALYSIS RESULTS

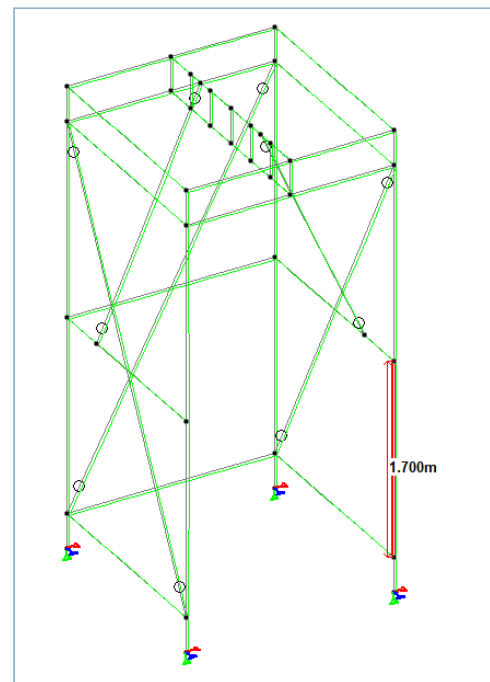
Utilization Ratio (UC) Diagram



Vertical Deflection Diagram



Horizontal Deflection Diagram



APPENDIX D - GUI RESULTS TABLES

Member Force / Code Check Table (Connection Stability)

Beam	L/C	Node	Axial Force kN	Shear-Y kN	Shear-Z kN	Torsion kN-m	Moment-Y kN-m	Moment-Z kN-m
96	16	45	4.109	-0.947	-0.087	-0.006	0.007	-0.194
96	20	45	3.621	-0.878	-0.087	-0.006	0.008	-0.183
96	14	45	3.599	-0.896	-0.088	-0.006	0.007	-0.186
75	20	48	3.596	0.891	0.033	0.006	0.009	-0.085
103	16	36	3.592	-0.911	0.031	0.006	-0.007	0.184
75	14	48	3.585	0.907	0.033	0.006	0.009	-0.087
104	14	51	3.545	0.964	-0.090	-0.006	0.028	0.093
75	16	48	3.542	0.958	0.033	0.006	0.009	-0.094
103	14	36	3.531	-0.962	0.031	0.006	-0.007	0.191
104	20	51	3.522	0.982	-0.090	-0.006	0.028	0.096
103	20	36	3.520	-0.978	0.030	0.006	-0.008	0.194
104	15	51	3.413	0.883	0.412	0.008	0.046	0.081
103	15	36	3.397	-0.875	-0.146	-0.007	-0.097	0.179
96	15	45	3.319	-0.965	-0.483	-0.024	0.179	-0.195
75	15	48	3.303	1.006	0.066	0.024	-0.068	-0.106
96	19	45	3.131	-0.879	-0.002	-0.011	0.027	-0.182
75	19	48	3.110	0.916	-0.072	0.011	-0.040	-0.092
104	19	51	3.069	0.954	0.029	-0.011	0.010	0.094
104	16	51	3.052	0.914	-0.089	-0.006	0.027	0.086
103	19	36	3.047	-0.980	-0.109	0.011	-0.027	0.190
101	16	49	2.557	-1.061	-0.115	-0.002	-0.014	-0.144
109	16	54	2.557	-0.665	0.088	-0.002	-0.031	0.016
101	20	49	2.210	-0.987	-0.115	-0.002	-0.013	-0.134
109	20	54	2.210	-0.600	0.084	-0.002	-0.030	0.015
100	20	46	2.162	0.821	0.005	0.002	0.016	-0.118
106	16	50	2.161	-0.776	-0.010	-0.000	-0.021	0.119
100	14	51	2.147	0.814	0.007	0.002	0.014	-0.116

Vertical Displacement Table (SLS Combinations)

Beam	Max Disp mm	Location m	L/C	L/Displ	Global X mm	Global Y mm	Global Z mm
28	2.851	1.050	17	631	52.285	-0.011	7.011
28	2.806	1.050	19	641	51.620	0.097	6.862
24	2.797	0.750	17	643	53.006	-0.023	-7.010
24	2.675	0.750	19	672	53.348	0.056	-6.869
28	1.376	1.050	15	1308	20.690	0.009	3.436
121	1.230	0.450	19	1463	53.738	-0.648	4.470
24	1.207	0.750	15	1491	22.386	0.069	-3.422
121	1.130	0.450	17	1593	53.951	-0.715	4.373
120	0.909	0.750	19	1979	56.605	-0.127	2.025
120	0.771	0.900	15	2334	22.919	-0.122	0.789
45	0.761	0.900	19	1773	64.920	-1.106	-6.719
121	0.733	0.450	15	2456	22.301	-0.244	2.411
45	0.646	0.787	17	2088	63.609	-0.547	-6.857
45	0.626	0.900	15	2157	28.140	-1.247	-3.331
49	0.611	0.787	19	2209	61.219	-1.214	6.718
120	0.601	0.900	16	2995	-1.001	-0.082	19.601
120	0.584	0.900	20	3080	-1.042	-0.079	35.362
120	0.573	0.900	14	3141	-1.012	-0.093	0.558
49	0.565	0.675	17	2388	63.699	-0.473	6.878
49	0.477	0.900	20	2832	-2.102	-0.924	34.616
49	0.457	0.900	14	2957	-2.065	-0.924	-0.182
49	0.453	0.900	16	2982	-2.087	-0.901	18.835
120	0.403	0.750	17	4468	57.569	-0.110	1.548
24	0.377	0.450	18	4779	0.021	-0.377	32.637
28	0.372	1.350	18	4842	-0.034	-0.372	32.636
66	0.355	1.350	17	5069	68.245	-0.062	-7.221

Max = 1.076 mm (LC 16) | L = 3000 mm | Allowable L/100 = 30.0 mm -> **PASS**

Horizontal Displacement Table (SLS Combinations)

Beam	Max Disp mm	Location m	L/C	L/Displ	Global X mm	Global Y mm	Global Z mm
38	2.809	0.992	19	605	63.101	-0.004	-6.401
38	2.453	0.992	17	693	61.200	0.008	-6.749
40	2.045	0.992	17	831	58.580	-0.009	7.066
40	1.852	0.992	19	917	56.748	-0.019	7.068
61	1.719	0.850	17	988	70.947	-0.012	6.904
61	1.686	0.850	19	1008	69.510	-0.035	6.612
59	1.611	0.708	19	1055	72.359	-0.004	-7.151
59	1.600	0.708	17	1062	71.743	0.021	-7.126
38	1.438	0.992	15	1182	27.877	-0.015	-2.913
36	0.951	0.850	17	1787	56.698	-0.050	-7.054
34	0.940	0.850	19	1807	56.009	0.005	6.358
36	0.930	0.850	19	1827	56.066	-0.059	-7.033
34	0.922	0.850	17	1843	56.671	0.017	6.788
61	0.881	0.992	15	1929	29.590	-0.029	3.330
59	0.872	0.567	15	1950	32.107	-0.024	-3.780
38	0.829	0.992	20	2049	2.029	-0.036	34.793
40	0.728	0.992	20	2336	-2.030	-0.020	34.421
38	0.543	0.992	18	3128	0.128	-0.023	34.444
38	0.540	0.992	16	3150	2.138	-0.035	19.419
55	0.536	0.567	20	3169	-0.864	-0.005	34.477
40	0.534	0.992	18	3185	-0.198	-0.009	34.419
36	0.504	0.992	18	3373	-0.064	-0.004	34.407
34	0.478	0.992	18	3559	-0.053	0.009	34.361
55	0.472	0.567	19	3602	56.114	0.003	7.225
40	0.434	0.992	15	3913	21.504	-0.015	3.549
38	0.428	0.992	14	3967	2.037	-0.020	0.393

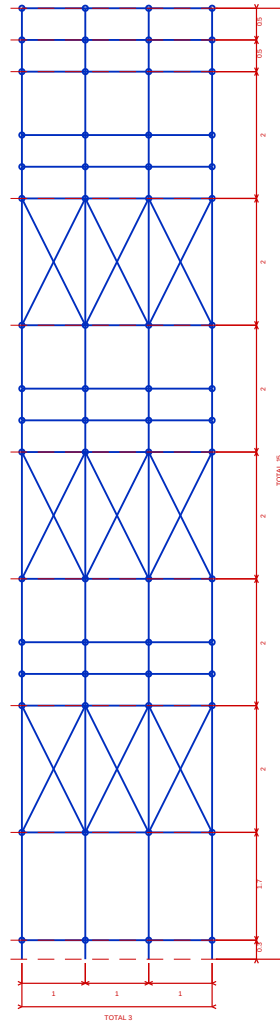
Max = **14.114 mm** (LC 16) | L = 4000 mm | Allowable L/200 = 20.0 mm -> **PASS**

APPENDIX E - REFERENCES & STANDARDS

Reference	Title	Year
BS EN 12811-1:2003	Temporary Works Equipment - Scaffolds - Performance Requirements and General Design	2003
BS EN 1993-1-1:2005	Eurocode 3: Design of Steel Structures - General Rules and Rules for Buildings	2005
BS EN 1991-1-4:2005	Eurocode 1: Actions on Structures - Wind Actions	2005
BS EN 1990:2002	Eurocode: Basis of Structural Design	2002
NASC TG20:13	Guide to Good Practice for Scaffolding with Tubes and Fittings	2013
Dredger P.	How to Approximate 10-Minute Mean Wind Speed from 3-Second Gust	-
STAAD.Pro CONNECT	STAAD.Pro CONNECT Edition V22 - Structural Analysis and Design Software	2022

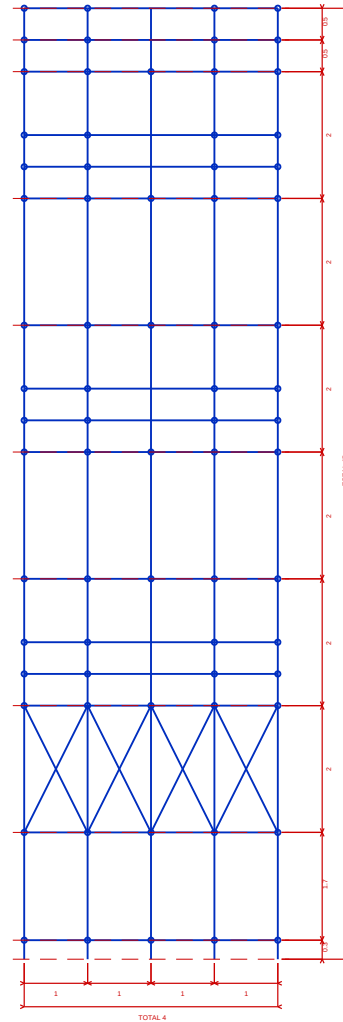
Software Used: STAAD.Pro CONNECT Edition V22 (Bentley Systems) - 3D Structural Analysis and Code Checking per EN 1993-1-1:2005.

Report: Scaffold Structural Design Report - ARCO M&E, NLNG Bonny Island.



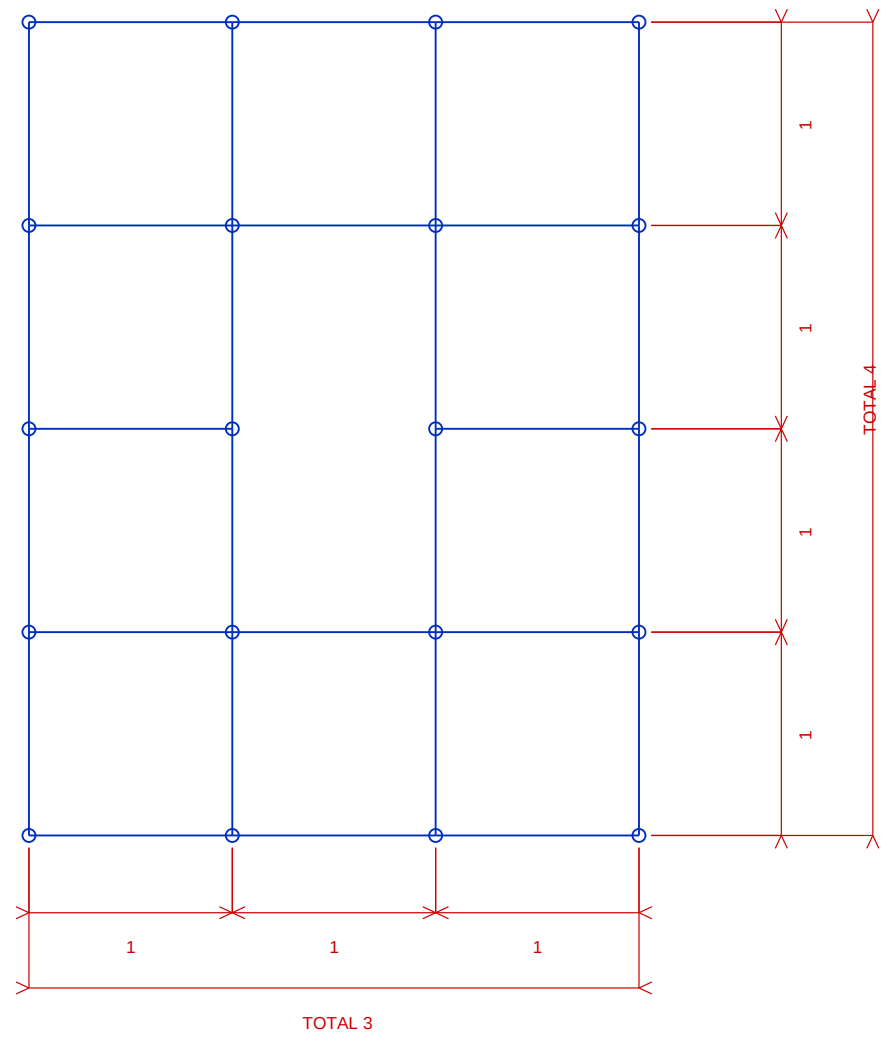
FRONT VIEW

PROJECT		DRAWING NO.			 NLNG	 ARCO M&E
SCAFFOLDING WORKS		PMS-WA-ARCO-003-FRONT-VIEW				
DRAWING TITLE		SCALE	DATE	REV	COMPANY	
SCAFFOLD GENERAL ARRANGEMENT - FRONT VIEW		NTS	14-JUN-2026	0		
CLIENT	LOCATION	DRAWN	CHECKED	APPROVED	SHEET	
		Nnamani, Ifeanyi				
					1 OF 4	A4



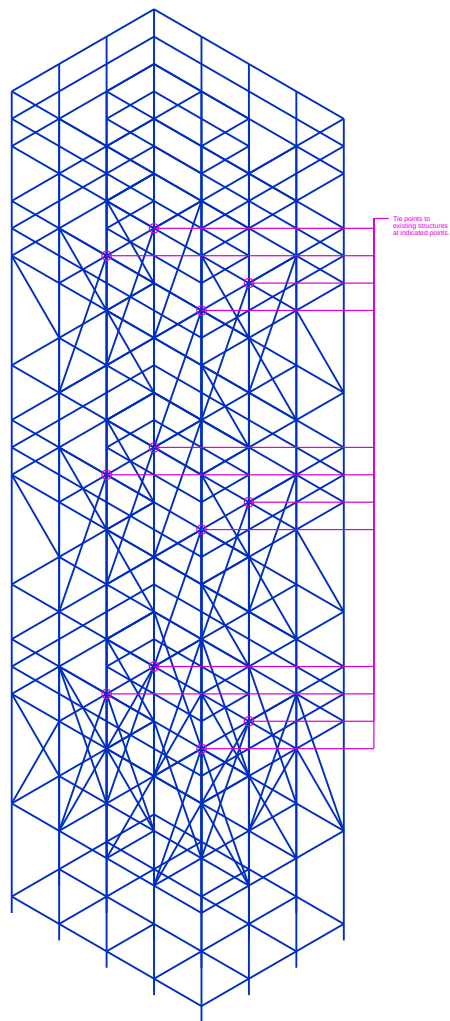
SIDE VIEW

PROJECT SCAFFOLDING WORKS		DRAWING NO. PMS-WA-ARCO-003-SIDE-VIEW			 NLNG			
DRAWING TITLE SCAFFOLD GENERAL ARRANGEMENT - SIDE VIEW		SCALE NTS	DATE 14-JUN-2026		REV 0		COMPANY	
CLIENT	LOCATION	DRAWN Nnamani, Ifeanyi	CHECKED	APPROVED		SHEET 2 OF 4		SIZE A4



PLAN VIEW

PROJECT SCAFFOLDING WORKS		DRAWING NO. PMS-WA-ARCO-003-PLAN-VIEW			 NLNG	
DRAWING TITLE SCAFFOLD GENERAL ARRANGEMENT - PLAN VIEW		SCALE NTS	DATE 14-JUN-2026	REV 0		
CLIENT		LOCATION	DRAWN Nnamani, Ifeanyi	CHECKED	APPROVED	COMPANY
		</				



3D VIEW

PROJECT		DRAWING NO.						
SCAFFOLDING WORKS		PMS-WA-ARCO-003-3D-VIEW						
DRAWING TITLE		SCALE	DATE	REV				
SCAFFOLD GENERAL ARRANGEMENT - 3D VIEW		NTS	14-JUN-2026	0				
CLIENT		LOCATION		DRAWN	CHECKED	APPROVED		
				Nnamani, Ifeanyi				
						SHEET		SIZE
						4 OF 4		A4

3D Isometric View

